

Runwise Developer Guide

Stack

Prerequisites

- Node.js 22 or later
 - npm
-

Setup

```
git clone https://github.com/alanwaddington/runwise.git
cd runwise
npm install
npm run dev
```

Open http://localhost:5173.

Project Structure

```
src/
% % % app.css          # Global styles + Tailwind theme tokens
% % % app.html         # SvelteKit HTML shell
% % % lib/
%   % % % components/  # Shared UI components
%   %   % % % HeroSection.svelte
%   %   % % % HeroSection.test.ts
%   %   % % % InputField.svelte
%   %   % % % InputField.test.ts
%   %   % % % ResultDisplay.svelte
%   %   % % % ResultDisplay.test.ts
%   %   % % % SiteNav.svelte
%   %   % % % SiteNav.test.ts
%   %   % % % ToolCard.svelte
%   %   % % % ToolCard.test.ts
%   %   % % % ToolLayout.svelte
%   %   % % % ToolLayout.test.ts
%   % % % utils/        # Pure utility modules (no Svelte dependency)
%   %   % % % pace.ts    # Pace/speed conversion functions
%   %   % % % pace.test.ts
%   %   % % % race-predictor.ts  # Riegel formula, time parsing/formatting, prediction
table
%   %   % % % race-predictor.test.ts
%   %   % % % training-paces.ts  # VDOT calculation (Daniels' formula), training zone
pace derivation
%   %   % % % training-paces.test.ts
%   %   % % % hr-zones.ts      # Max HR zones, Friel LTHR zones, LTHR sub-zones,
Tanaka age estimate
%   %   % % % hr-zones.test.ts
%   %   % % % vo2max.ts        # ACSM normative data, getFitnessCategory,
getAcsmTable, CATEGORY_COLOURS
%   %   % % % vo2max.test.ts
%   %   % % % parkrun.ts       # Effort mapping, Riegel prediction, split generation,
PB comparison, WMA age grading
%   %   % % % parkrun.test.ts
% % % routes/
%   % % % +layout.svelte      # Root layout – header + main wrapper
%   % % % +page.svelte        # Home page – HeroSection + ToolCard grid
```

```
% % % +error.svelte      # Error page
% % % pace/
% % % race-predictor/
% % % training-paces/
% % % hr-zones/
% % % vo2max/
% % % parkrun/
```

Design System

Tokens (`src/app.css`)

Dark mode is applied automatically via prefers-color-scheme. Always use design tokens (text-ink, bg-bg, border-ink/10) rather than hardcoded Tailwind colours.

Components

Adding a New Tool

- Create `src/routes/<tool-name>/+page.svelte`
- Import `ToolLayout` and pass `title` and `description` props
- Add the tool to the `tools` array in `src/routes/+page.svelte`
- Add the tool link to `SiteNav.svelte`
- Register the route in `src/lib/seo.ts` (`PAGES` map) with a title, description, OG image path, and sitemap priority — this drives the page's meta tags, `sitemap.xml` entry, `robots.txt` allow rules, and JSON-LD structured data
- Add tests alongside the page

Testing

Unit and component tests (Vitest)

```
npm run test          # Run all tests once
npm run test -- --watch # Watch mode
```

Tests live alongside their source files (*.test.ts). Follow the existing pattern:

```
import { describe, it, expect, afterEach } from 'vitest';
import { render, cleanup, screen } from '@testing-library/svelte';
import MyComponent from './MyComponent.svelte';

afterEach(() => { cleanup(); });

describe('MyComponent', () => {
  it('renders correctly', () => {
    render(MyComponent, { props: { ... } });
    expect(screen.getByRole('...')).toBeInTheDocument();
  });
});
```

E2E tests (Playwright)

```
npx playwright install chromium # First time only
npm run test:e2e                # Run E2E tests against the production build
```

E2E tests live in the e2e/ directory. They run against the production preview server (npm run build && npm run preview on port 4173). Configuration is in playwright.config.ts.

Code Quality

```
npm run check          # TypeScript / svelte-check
```

```
npm run lint      # ESLint
npm run format    # Prettier (auto-fix)
npm run test:e2e  # Playwright E2E tests (requires built app)
```

Workflow

This project follows an issue-driven workflow:

```
/analyse <issue>    !' requirements + acceptance criteria
/design <issue>      !' architecture + work breakdown
/develop <issue>     !' TDD implementation
/verify <PR>         !' runtime verification
/pr-reviewer <PR>    !' acceptance criteria audit
/merge <PR>          !' merge + close issues
```